Dated class-number bounds

28 October 2026 UTC | T-5518 | source audit; no new scientific trials

Documented h(K) >= 200	Documented h(K) > 10^7
10 published / 11 omitted PASS*	NOT EXECUTED: certificate planned

*Conditional on the external h <= 2000 list completeness theorem

Completed comparison: the dated lower bound200 passes all ten publications and eleven omissions, conditional on the existing external list completeness theorem. **The stronger10^7 condition has not been evaluated.**

Source dates. Baier2002 section2.3.5 reports h(O_K)>=200. ETSI TS102176-1 V1.2.1 (July2005), clause6.1.2.3 and AnnexD.3, uses200. RFC5639 (March2010), section2.1 item3, requires h(O_K)>10^7. All concern the maximal CM order. No1997 NIST adoption is established.

Evidence limit. The earlier full21 study found no field on an external list claiming completeness for h(K)<=2000. The new crosswalk binds that analysis and each UID. Conditional on the theorem, h(K)>2000 implies the documented200 condition. It does not establish h(K)>10^7. This session did not independently prove the external classification.

Order distinction. The class number of the maximal order divides that of any order in the field. A certified maximal-order lower bound therefore also applies to the actual CM order. A Frobenius-order lower bound alone cannot be transferred in the reverse direction. Conductor3 alone does not imply small class number.

Prospective certificate. For24 distinct split primes with4*product<|D_K|, the subset-ideal argument in CERTIFICATE-LEMMA.md proves h(K)>=2^24>10^7. PLAN.md fixes ascending-prime selection and UNKNOWN/NOT_CERTIFIED outcomes. Implementation, independent arithmetic checks and formal protocol freezing remain required before new cohort evaluation.

Primary sources. d-nb.info/964515040/34; ETSI TS102176-1 V1.2.1 July2005; rfc-editor.org/rfc/rfc5639.html; arxiv.org/abs/2610.21511. Exact source URLs and prior-input digest are in SOURCE-AUDIT.json. Full report.md and editable figure generator accompany this digest. Historical selector unresolved; goal active.